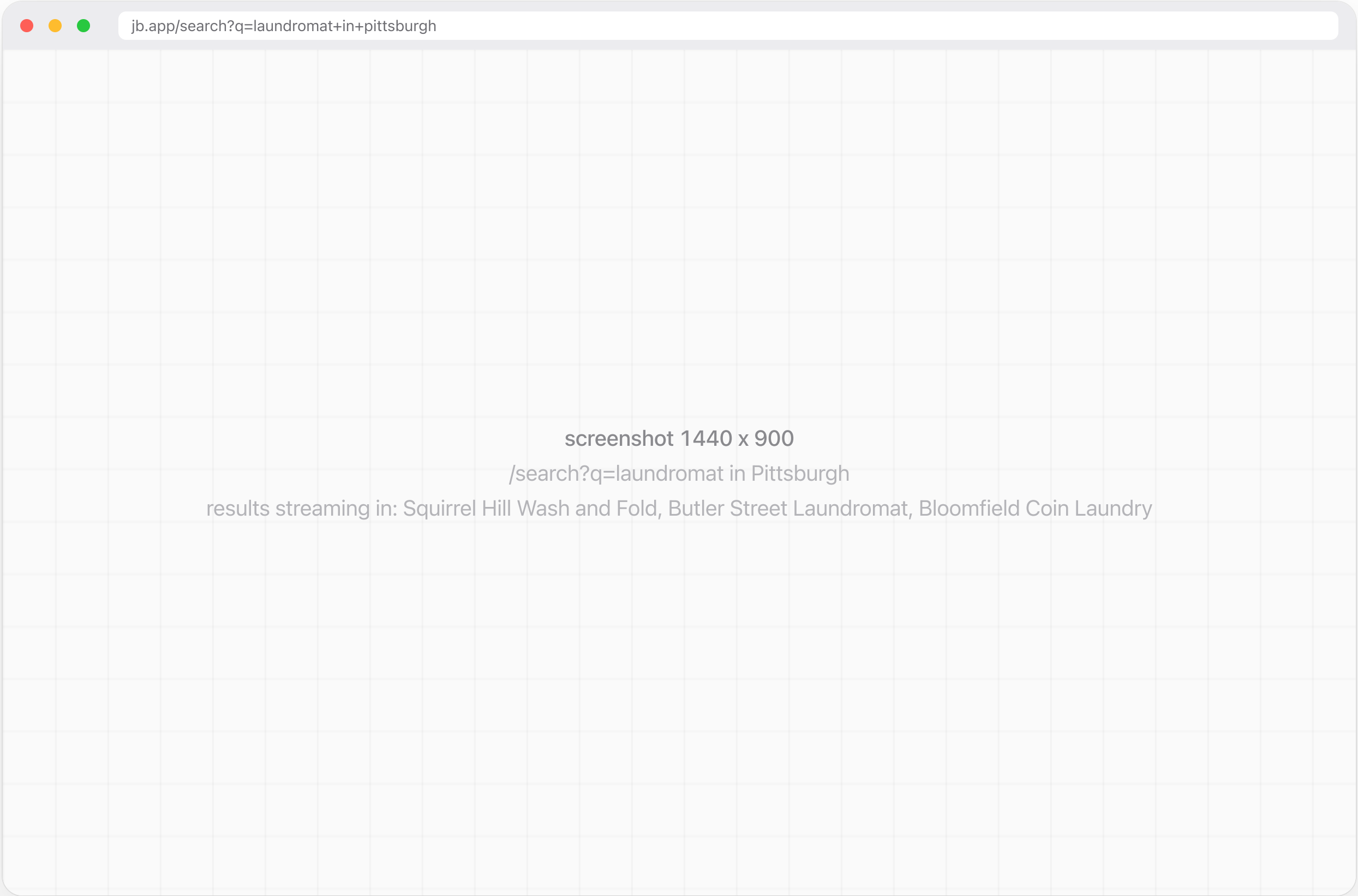


JB

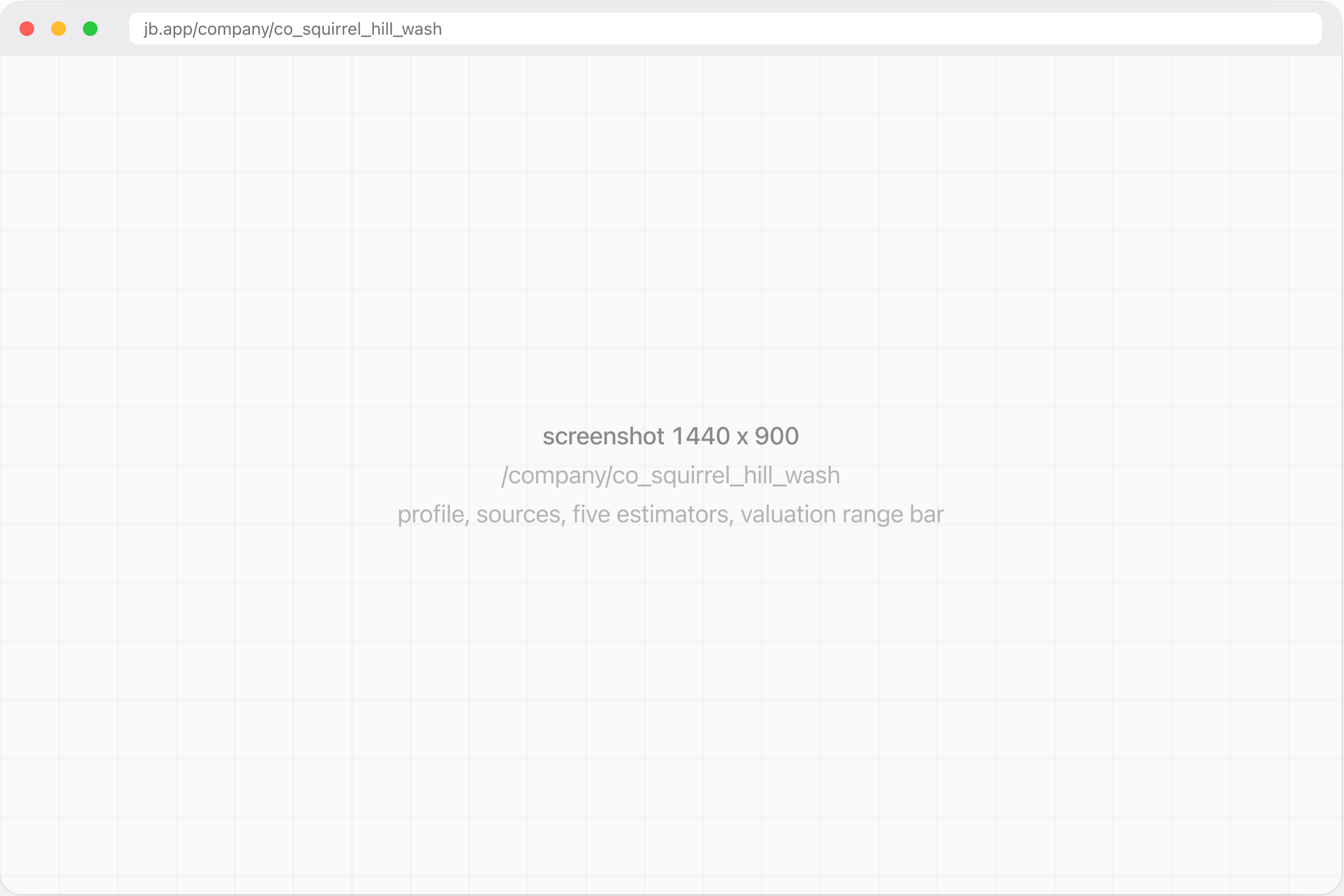
A stock market for the businesses that will never be listed.

33 million small businesses. None of them has a price.

If you wanted to buy a laundromat in Squirrel Hill tonight, you could not even find the list.



One company



Squirrel Hill Wash and Fold

\$570,768

\$475k to \$686k · σ 0.22

\$57.08 a share

How we price it

$$\ln V \mid k \sim \mathcal{N}(\mu_k, \sigma_k^2)$$

$$\mu_{\text{list}} = \ln(h \cdot P_{\text{ask}}), \quad \sigma_{\text{list}} = 0.15$$

$$\mu_{\text{inc}} = \ln\left(m_c \cdot h \cdot \text{SDE} \cdot \prod_j q_j\right), \quad \sigma_{\text{inc}} \in \{0.25, 0.40\}$$

$$\text{SDE} = R \cdot \phi_c \text{ if unobserved}$$

$$\mu_{\text{proxy}} = \ln\left(m_c \cdot h \cdot \phi_c \cdot \hat{R}\right), \quad \hat{R} = \frac{1}{|J|} \sum_{j \in J} n_j \rho_j, \quad \sigma_{\text{proxy}} = 0.60$$

$$\mu_{\text{llm}} = \frac{1}{M} \sum_{m=1}^M \ln \hat{V}_m$$

$$\sigma_{\text{llm}} = \max\left(0.35 + 0.35(1 - c), \frac{1}{2} \mid \ln \hat{V}_1 - \ln \hat{V}_2 \mid\right)$$

$$\mu_{\text{base}} = \ln\left(\tilde{P}_c \cdot h \cdot \kappa_s \cdot \prod_j q_j\right), \quad \sigma_{\text{base}} = 0.70$$

$h = 0.88$ sold to asking haircut $\cdot m_c$ category SDE multiple (BizBuySell 2026) $\cdot \phi_c$ category SDE margin $\cdot q_j$ quality multipliers (reviews, tenure, owner operated) $\cdot \tilde{P}_c$ category median asking price $\cdot \kappa_s$ state price index $\cdot n_j, \rho_j$ observable counts and revenue per unit (employees, machines, reviews) $\cdot \hat{V}_m$ model opinions (Grok, K2) $\cdot c$ model confidence $\cdot \Phi^{-1}$ inverse normal CDF $\cdot \hat{\sigma}_k$ calibration on scraped listings with the price hidden

Precision weighted Bayesian ensemble in log space

$$w_k = \sigma_k^{-2}, \quad \mu = \frac{\sum_k w_k \mu_k}{\sum_k w_k}$$

$$\sigma^2 = \frac{1}{\sum_k w_k} + \frac{\sum_k w_k (\mu_k - \mu)^2}{\sum_k w_k}$$

$$\sigma \leftarrow \text{clip}(\sigma, 0.12, 0.90), \quad V_0 = e^\mu$$

$$V_q = \exp\left(\mu + \sigma \Phi^{-1}(q)\right)$$

$$\text{floor bid} = \frac{V_{0.20}}{N}, \quad N = 10,000$$

$$\text{ask ladder} = \left\{ \frac{V_q}{N} : q \in \{0.55, 0.6125, 0.675, 0.7375, 0.80\} \right\}$$

$$\hat{\sigma}_k = \text{std}_i(\mu_k^{(i)} - \ln(h \cdot P_{\text{ask}}^{(i)}))$$

$$D(p) = \sum_{b: \ell_b \geq p} q_b, \quad S(p) = \sum_{s: \ell_s \leq p} q_s$$

$$p^* = \arg \max_{p \in \mathcal{L}} \min\{D(p), S(p)\} \quad \text{then} \quad \arg \min |D(p) - S(p)| \quad \text{then} \quad \arg \min |p - p_{t-1}|$$

$$p^* \leftarrow \text{clip}(p^*, (1 - \beta) p_{t-1}, (1 + \beta) p_{t-1}), \quad \beta = 0.10$$

$$x_i = q_i \cdot \frac{\min\{D(p^*), S(p^*)\}}{D(p^*)} \text{ (buyers, if } D > S), \quad \text{everyone trades at } p^*$$

$$T = \text{clip}\left(\frac{200}{n_{\text{open}}}, 10, 60\right) \text{ seconds}$$

$$w_t = \frac{v_t}{\bar{v}}, \quad \mu_{\text{post}} = \frac{\mu/\sigma^2 + \sum_t w_t \ln(Np_t^*)/s_m^2}{1/\sigma^2 + \sum_t w_t/s_m^2}, \quad \sigma_{\text{post}}^2 = \frac{1}{1/\sigma^2 + \sum_t w_t/s_m^2}$$

$$\forall u : \{b \in B_u, s \in S_u, \ell_b \geq \ell_s\} \Rightarrow \text{cancel } \arg \max(\text{seq}), \quad q \leq 0.05N, \quad \text{notional}_u \leq 0.25 \text{ cash}_u$$

ℓ limit price · q quantity · $\mathcal{L}$ set of limit prices in the book · p_{t-1} last clearing price · β price band · n_{open} open orders · v_t round volume, $\bar{v}$ average volume · s_m per round price noise (0.10, then realized) · μ, σ valuation posterior from the previous slide · $N = 10,000$ shares · B_u, S_u one user's bids and asks · the platform never trades

Live

Squirrel Hill Wash and Fold · round 2

Bids			qty · limit	Asks			qty · limit
180.00	64.20	180.00 filled		284.64	58.67	105.15 filled	
20.00	62.00	20.00 filled		600.00	60.76	221.66 filled	
92.50	61.10	92.50 filled		600.00	63.03		owner
34.31	60.80	34.31 filled		600.00	65.59		owner
60.00	54.80		bot	600.00	68.62		owner
55.00	54.60		bot				
48.00	54.30		bot				
25.00	49.10		bot				
1,000.00	47.48		owner floor				

60.76

clearing price



QR
scan and bid

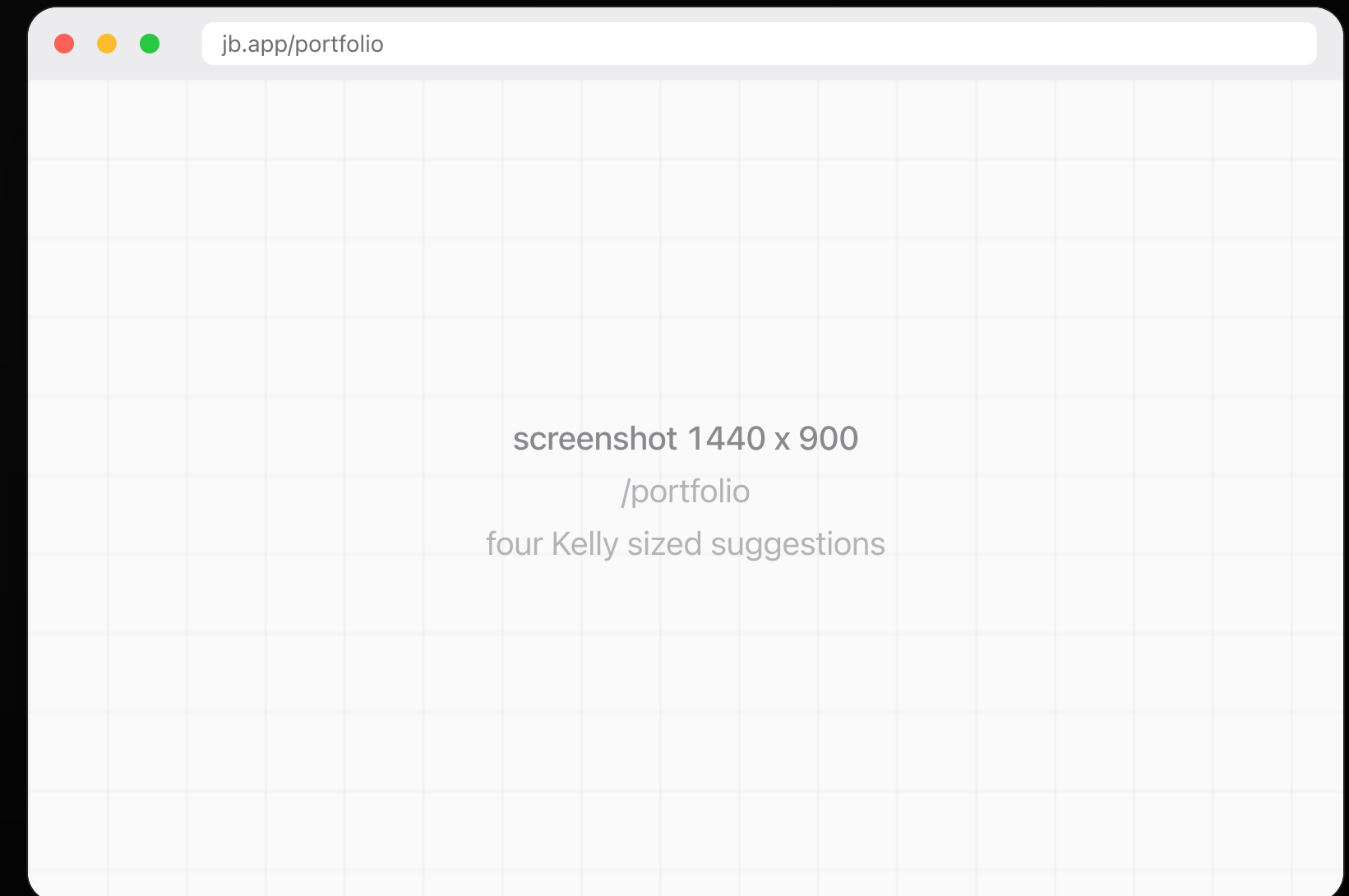
326.81 shares traded at 60.76 · 4 bids filled · owner sold 326.81 from the ladder · next batch in 10

Portfolio

$$\mu = \ln \frac{v}{p}, \quad f^* = \frac{\mu}{\sigma^2}$$
$$f = \text{clip}\left(\frac{1}{2} f^*, 0, 0.20\right), \quad \$ = f \cdot B$$

Sized by half Kelly on the gap between model value and market price.

v model value per share · p market price · σ posterior uncertainty · B bankroll · half Kelly because the edge is estimated



Acquire

From a share to the whole company.

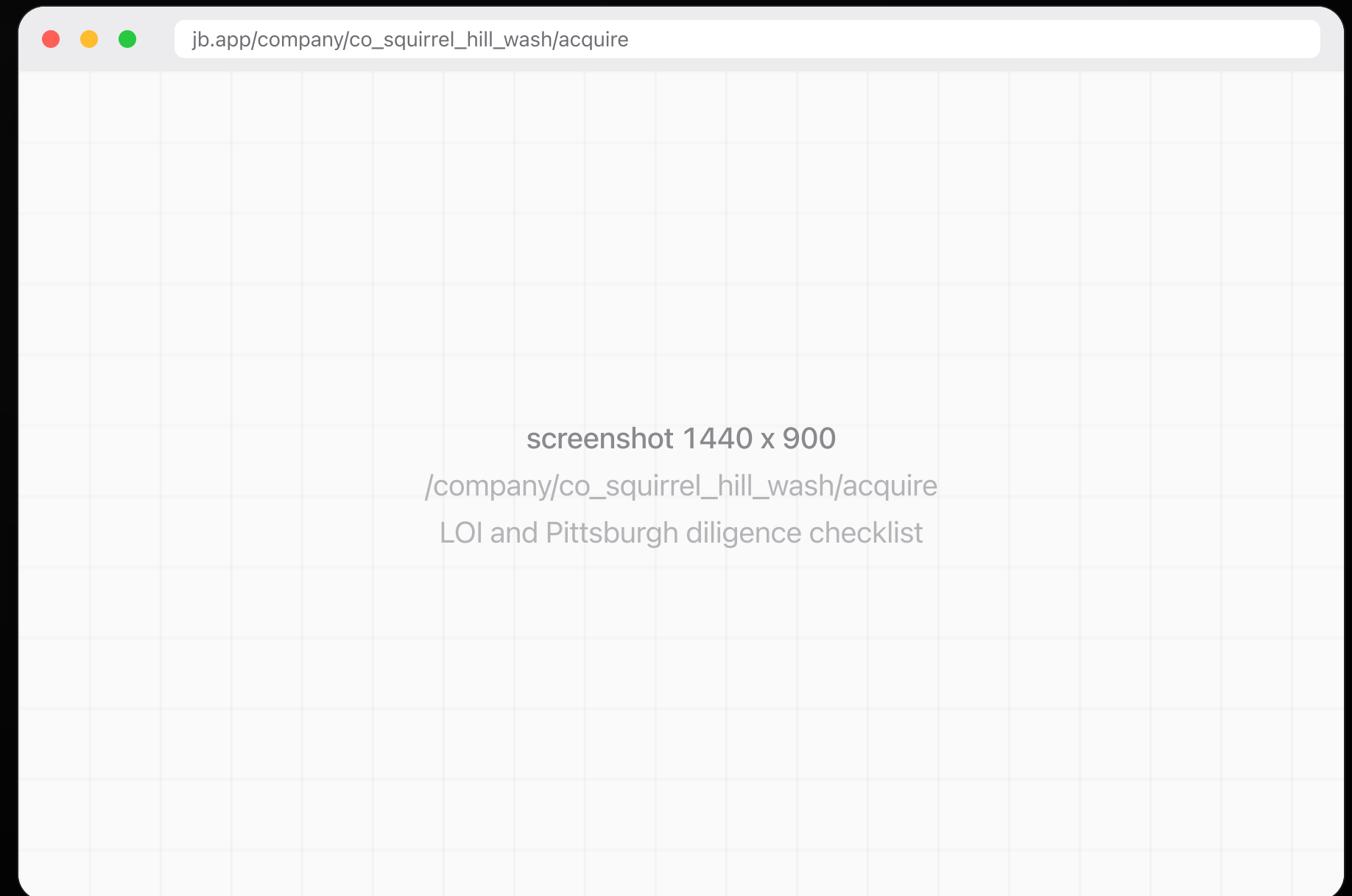
Letter of intent, drafted

City of Pittsburgh business registration

Allegheny County Health Department permit

PA bulk sale clearance

Every item cited



Every round, two model families read the tape.

● **Flagged** wash trade · round 14 · bot_17 and bot_23

Grok: the two accounts have traded only with each other for 9 rounds, alternating sides at the clearing price. No economic purpose.

K2 concurs independent review

Same pair, same pattern, 0.94 confidence. Orders cancelled, accounts frozen, entry written to the audit log.

Wash: same user on both sides within 3 rounds, or two accounts that only ever trade with each other. · Pump: $|\Delta \ln p^*| > 2\sigma_p$ with more than 60% of buy volume from one account.

Stack

Grok

xAI

IFM K2

Horizon

Querit

web search

MongoDB Atlas

Vector Search

Auth0

login

Vultr

API

Vercel

web

built in Cursor

Next.js 15 · FastAPI · Python · track: Optimization

JB. Price everything.

Nico Vir Aditya Zhiyuan

QR
live site